

EXP 7

Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM

To create a Servlet-based web application that stores user preferences (such as theme or language) using cookies and retrieves them on subsequent visits.

ALGORITHM

1. Start the application.
2. Accept user preferences (theme/language) through a form.
3. Create cookies for each preference.
4. Set the cookie expiry time.
5. Store cookies in the client's browser.
6. When the user revisits the application:
 - Read cookies from the request.
 - Check for stored preferences.
7. If cookies are found:
 - Retrieve and display the stored preferences.
8. If cookies are not found:
 - Display default values or ask the user to set preferences.
9. Stop the application.

STEPS

1. Design a user interface to collect preferences like theme and language.
2. Configure a Servlet to handle form submission.
3. Use cookies to store the user's preferences in the browser.
4. Set an appropriate expiry time for cookies.
5. Send cookies along with the response to the client.
6. On subsequent visits, retrieve cookies from the request object.
7. Extract stored values (theme/language).
8. Display the preferences to the user dynamically.

9. Handle cases where cookies are absent by showing default settings.

Program:

preferences.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Preferences</title>

  <style>
    body{
      font-family: Arial;
      background:#dbeafe;
      text-align:center;
      margin-top:100px;
    }

    .box{
      background:white;
      width:400px;
      margin:auto;
      padding:25px;
      border-radius:10px;
      box-shadow:0 0 10px gray;
    }

    select, button{
      width:90%;
      padding:10px;
      margin:10px;
    }

    button{
      background:#0d47a1;
      color:white;
      border:none;
    }
```

```
</style>
</head>
<body>

<div class="box">

<h2>Student Preferences</h2>

<form action="preferences" method="post">

    <select name="theme">
        <option>Light</option>
        <option>Dark</option>
    </select>

    <select name="language">
        <option>English</option>
        <option>Tamil</option>
    </select>

    <button type="submit">Save Preferences</button>

</form>

</div>

</body>
</html>
```

PreferenceServlet.java:

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/preferences")
public class PreferenceServlet extends HttpServlet {
```

```

protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    String theme = request.getParameter("theme");
    String language = request.getParameter("language");

    Cookie c1 = new Cookie("theme", theme);
    Cookie c2 = new Cookie("language", language);

    c1.setMaxAge(60*60*24);
    c2.setMaxAge(60*60*24);

    response.addCookie(c1);
    response.addCookie(c2);

    response.sendRedirect("preferences");
}

```

```

protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    String theme = "Not Set";
    String language = "Not Set";

    Cookie cookies[] = request.getCookies();

    if(cookies != null){
        for(Cookie c : cookies){

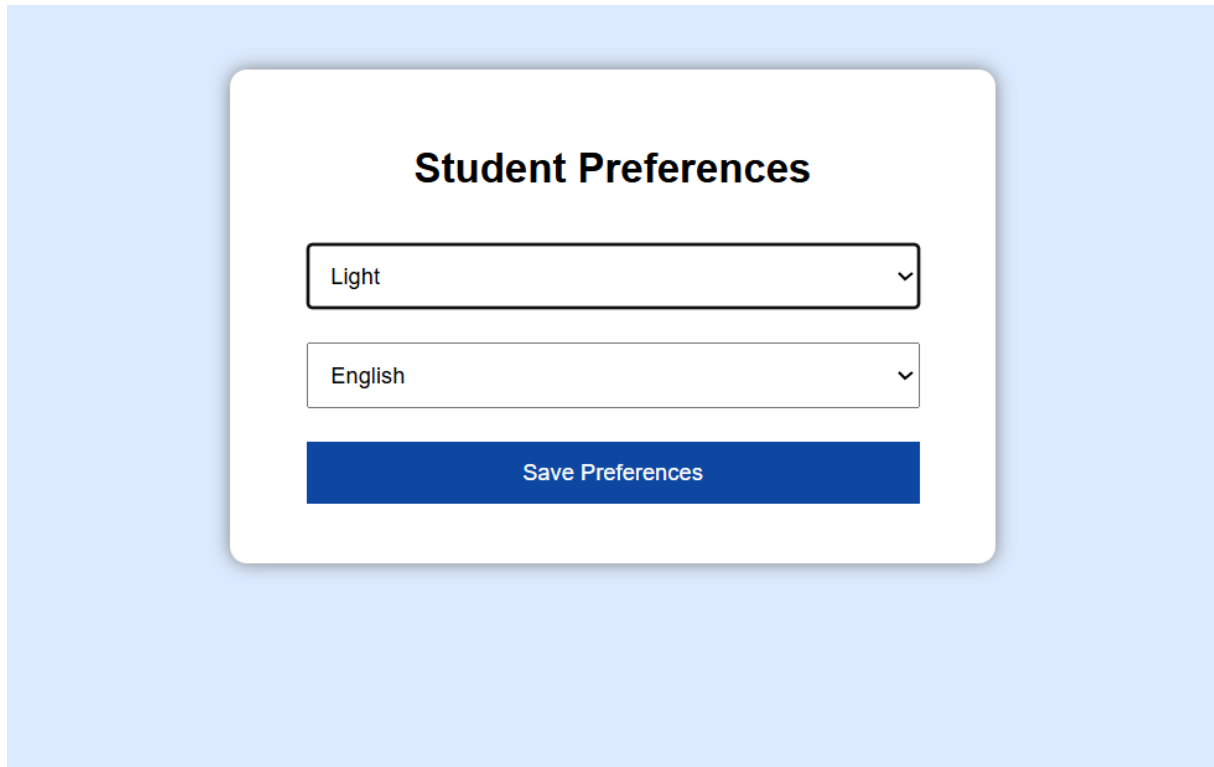
            if(c.getName().equals("theme")){
                theme = c.getValue();
            }

            if(c.getName().equals("language")){
                language = c.getValue();
            }
        }
    }
}

```

```
    }  
}  
  
out.println("<html><body style='font-family:Arial;text-align:center;margin-top:100px;'>");  
  
out.println("<h1>Student Information Management System</h1>");  
out.println("<h2>Saved Preferences</h2>");  
  
out.println("<p>Theme: " + theme + "</p>");  
out.println("<p>Language: " + language + "</p>");  
  
out.println("<a href='preferences.html'>Change Preferences</a>");  
  
out.println("</body></html>");  
}
```

OUTPUT:

A screenshot of a web form titled "Student Preferences". The form is centered on a light blue background. It contains two dropdown menus: the first is set to "Light" and the second is set to "English". Below these is a blue button labeled "Save Preferences".

Student Preferences

Light

English

Save Preferences

Student Information Management System

Saved Preferences

Theme: Dark

Language: English

[Change Preferences](#)

RESULT:

The Servlet application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.

